Annular structure forced by recurrent transverse-path accumulation (Proposition 3.3)

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Editorial note. Only whole independent Proposition3.3 counted; all support and applications count zero.

Editorial note. Author native TikZ and mathematical text retained; forced publisher page breaks suppressed by the wrapper.

Editorial note. Original native lines 637 and 642 use the notation tilde dom(I) for a lifted codomain, whereas the section 3 setup uses tilde Sigma. This source notation is retained unchanged; no extra isotopy hypothesis or mathematical correction is inserted.

Editorial scope. Whole original Proposition3.3 once. Original math and figures preserved; independent wrapper, explicit counters and cited bibliography reconstruction only.

Allowed standard prerequisites (editorial).

- Jordan separation and elementary covering theory of oriented surfaces. Original source section2.1 and section3.1; reproduced conventions. Separates proper lifted leaves and interprets deck transforms and simple loops. The nonroutine recurrence/accumulation construction is retained.
- Elementary strip endpoint classification and transverse-path terminology. Original section3.1–3.2, including author Lemma3.1. Definitions and the source four alternatives retained as written; no generated proof. The counted independent Proposition3.3 has its full current nested proof.

Source reference. Original theorem 1.1: local-coloring-linear-logic/AHL237-annulus/sources/forward-reference.tex, source lines 1–5. Forward application pointer only. The geometric Proposition3.3 proof does not use the later area-preserving periodic-orbit theorem.

Source and counted claim

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Only whole original Proposition 3.3 (`prop: nontransitivity3`) is counted. It is the authors' independent strengthened accumulation result, not the later area-preserving periodic-orbit theorem. The surface is connected, oriented, boundaryless and not the torus, with the non-singular oriented foliation and transverse-path conventions of the source. All three conclusions, including the accumulated boundary leaf and its side condition, are retained.

Complete current core. The complete proof from the first paragraph through the final annulus and recurrence argument is preserved, including all nested lemmas, the full sublemma proof and all native TikZ figures. No support lemma is counted separately. The forward application pointer to a later dynamical strengthening is omitted as nondependency; the author's statement of Theorem 1.1 is retained only as evidence for another non-load-bearing forward reference.

Prerequisites and difficulty. Jordan separation and ordinary surface covering theory are standard inputs. The source's elementary strip terminology and four endpoint alternatives remain as written. The H2 core constructs a transverse loop from recurrence, controls competing deck translates through local leaf-order and return configurations, excludes access from the wrong side, and proves that the projected saturated strip is an annulus. It does not assume the later forcing or periodic-orbit theorem. No proof is generated or repaired and no mathematical correctness certification is supplied.

Identity boundary. Saved main and in-flight records have no exact DOI/arXiv/title hit. The geometric accumulation claim differs from mere existence of periodic points or homotopical rotation intervals. In-flight changes after 14 October 2026, 17:36:45 UTC remain unconfirmed; this is preparation only.

2 Definitions, notations and preliminaries

In the sequel, the letter S will refer to a closed surface while the letter Σ will refer to any surface (not necessarily compact, not necessarily connected). If f is a surface homeomorphism, μ will refer to any f-invariant measure, λ to an f-invariant measure with total support, and ν to an f-invariant ergodic measure.

2.1 Loops and paths

Let Σ be an oriented surface (not necessarily closed, not necessarily boundaryless, not necessarily connected). A *loop* of Σ is a continuous map $\Gamma : \mathbb{T} \to \Sigma$, where $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. It will be called *essential* if it is not homotopic to a constant loop. A *path* of Σ is a continuous map $\gamma : I \to \Sigma$ where $I \subset \mathbb{R}$ is an interval. A loop or a path will be called *simple* if it is injective. The *natural lift* of a loop $\Gamma : \mathbb{T} \to \Sigma$ is the path $\gamma : \mathbb{R} \to \Sigma$ such that $\gamma(t) = \Gamma(t + \mathbb{Z})$. A *segment* is a simple path $\sigma : [a, b] \to \Sigma$, where $a < b$. The points $\sigma(a)$ and $\sigma(b)$ are the *endpoints* of σ. We will say that σ *joins* $\sigma(a)$ to $\sigma(b)$. More generally if A and B are disjoint, we will say that σ joins A to B, if $\sigma(a) \in A$ and $\sigma(b) \in B$. A *line* is a proper simple path $\lambda : \mathbb{R} \to \Sigma$. As it is usually done we will use the same name and the same notation to refer to the image of a loop or a path γ.

Note that a simple loop or a simple path is naturally oriented by its parametrization (the increasing one). Let Γ be a simple loop of Σ, and denote Σ' the connected component of Σ it belongs to. If $\Sigma' \setminus \Gamma$ has two connected components, we say that Γ *separates* Σ; in this case the connected component that is located on the right of Γ will be denoted $R(\Gamma)$ and the other one $L(\Gamma)$. We will use the same notations $R(\lambda)$, $L(\lambda)$ for a line λ that separates the connected component it belongs to. In both cases, if γ is a segment that joins $R(\Gamma)$ to $L(\Gamma)$, then it holds that the algebraic intersection number between Γ and γ is 1.

Let f be an orientation preserving homeomorphism of Σ. A *Brouwer line* of f is a line λ that separates Σ such that $f(\lambda) \subset L(\lambda)$ and $f^{-1}(\lambda) \subset R(\lambda)$. Equivalently it means that $f(\overline{L(\lambda)}) \subset L(\lambda)$ or that $f^{-1}(\overline{R(\lambda)}) \subset R(\lambda)$.

3 Foliations on surfaces

In this section we will consider an oriented boundaryless surface Σ, not necessarily closed, not necessarily connected, and a non singular oriented topological foliation $\mathcal{F}$ on Σ. We will consider:

- the universal covering space $\widetilde{\Sigma}$ of Σ;
- the covering projection $\widetilde{\pi} : \widetilde{\Sigma} \to \Sigma$;
- the group $\mathcal{G}$ of covering automorphisms of this cover;
- the lifted foliation $\widetilde{\mathcal{F}}$ on $\widetilde{\Sigma}$.

For every point $z \in \Sigma$, we denote ϕ_z the leaf of $\mathcal{F}$ that contains z. If $\phi_z : \mathbb{R} \to \Sigma$ is a parametrization of ϕ_z inducing the orientation, such that $\phi_z(0) = z$, we set $\phi_z^+ = \phi_z|_{[0,+\infty)}$ and $\phi_z^- = \phi_z|_{(-\infty,0]}$. Similarly, for every point $\widetilde{z} \in \widetilde{\Sigma}$, we denote $\widetilde{\phi}_{\widetilde{z}}$ the leaf of $\widetilde{\mathcal{F}}$ that contains $\widetilde{z}$ and we define in the same way $\widetilde{\phi}_{\widetilde{z}}^+$ and $\widetilde{\phi}_{\widetilde{z}}^-$. The foliation $\widetilde{\mathcal{F}}$ being defined on a union of topological planes and being non singular, it has no closed leaf. Moreover a leaf cannot accumulate on a point, it must be proper. So it is a line and consequently it separates the connected component of $\widetilde{\Sigma}$ it belongs to (by Jordan theorem, because this connected component has to be a topological plane).

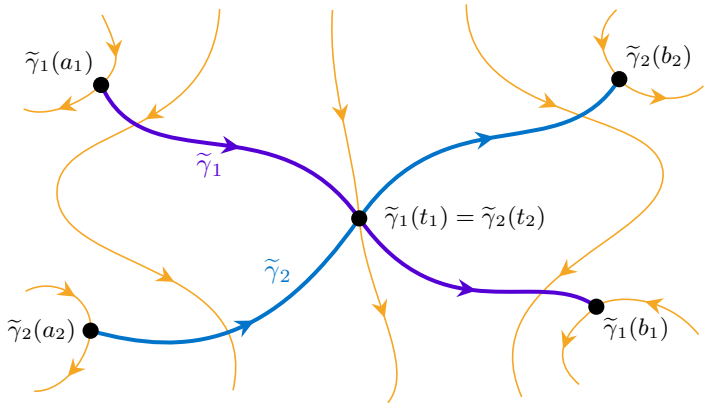


Figure 3.1: Example of $\widetilde{\mathcal{F}}$-transverse intersection.

3.1 $\mathcal{F}$-transverse intersections

A path $\gamma : J \to \Sigma$ is *positively transverse*[1] to $\mathcal{F}$ if it locally crosses each leaf of $\mathcal{F}$ from the right to the left. Observe that every lift $\widetilde{\gamma} : J \to \widetilde{\Sigma}$ of γ is positively transverse to $\widetilde{\mathcal{F}}$ and that for every $a < b$ in J:

- $\widetilde{\gamma}|_{[a,b]}$ meets once every leaf $\widetilde{\phi}$ such that $R(\widetilde{\phi}_{\widetilde{\gamma}(a)}) \subset R(\widetilde{\phi}) \subset R(\widetilde{\phi}_{\widetilde{\gamma}(b)})$;
- $\widetilde{\gamma}|_{[a,b]}$ does not meet any other leaf.

Two transverse paths $\widetilde{\gamma}_1 : J_1 \to \widetilde{\Sigma}$ and $\widetilde{\gamma}_2 : J_2 \to \widetilde{\Sigma}$ are said *equivalent* if they meet the same leaves of $\widetilde{\mathcal{F}}$. Two transverse paths $\gamma_1 : J_1 \to \Sigma$ and $\gamma_2 : J_2 \to \Sigma$ are *equivalent* if there exists a lift $\widetilde{\gamma}_1 : J_1 \to \widetilde{\Sigma}$ of γ and a lift $\widetilde{\gamma}_2 : J_2 \to \widetilde{\Sigma}$ of γ_2 that are equivalent.

Let $\widetilde{\gamma}_1 : J_1 \to \widetilde{\Sigma}$ and $\widetilde{\gamma}_2 : J_2 \to \widetilde{\Sigma}$ be two transverse paths such that there exist $t_1 \in J_1$ and $t_2 \in J_2$ satisfying $\widetilde{\gamma}_1(t_1) = \widetilde{\gamma}_2(t_2)$. We will say that $\widetilde{\gamma}_1$ and $\widetilde{\gamma}_2$ have a *$\widetilde{\mathcal{F}}$-transverse intersection* at $\widetilde{\gamma}_1(t_1) = \widetilde{\gamma}_2(t_2)$ if there exist $a_1, b_1 \in J_1$ satisfying $a_1 < t_1 < b_1$ and $a_2, b_2 \in J_2$ satisfying $a_2 < t_2 < b_2$ such that:

- $\widetilde{\phi}_{\widetilde{\gamma}_1(a_1)} \subset L(\widetilde{\phi}_{\widetilde{\gamma}_2(a_2)})$, $\widetilde{\phi}_{\widetilde{\gamma}_2(a_2)} \subset L(\widetilde{\phi}_{\widetilde{\gamma}_1(a_1)})$;
- $\widetilde{\phi}_{\widetilde{\gamma}_1(b_1)} \subset R(\widetilde{\phi}_{\widetilde{\gamma}_2(b_2)})$, $\widetilde{\phi}_{\widetilde{\gamma}_2(b_2)} \subset R(\widetilde{\phi}_{\widetilde{\gamma}_1(b_1)})$;
- every path joining $\widetilde{\phi}_{\widetilde{\gamma}_1(a_1)}$ to $\widetilde{\phi}_{\widetilde{\gamma}_1(b_1)}$ and every path joining $\widetilde{\phi}_{\widetilde{\gamma}_2(a_2)}$ to $\widetilde{\phi}_{\widetilde{\gamma}_2(b_2)}$ must intersect.

It means that there is a "crossing" between the two paths naturally defined by $\widetilde{\gamma}_1$ and $\widetilde{\gamma}_2$ in the space of leaves of $\widetilde{\mathcal{F}}$, which is a one-dimensional topological manifold, usually non Hausdorff (see Figure 3.1).

Now, let $\gamma_1 : J_1 \to \Sigma$ and $\gamma_2 : J_2 \to \Sigma$ be two transverse paths such that there exist $t_1 \in J_1$ and $t_2 \in J_2$ satisfying $\gamma_1(t_1) = \gamma_2(t_2)$. Say that γ_1 and γ_2 have a *$\mathcal{F}$-transverse intersection* at $\gamma_1(t_1) = \gamma_2(t_2)$ if $\widetilde{\gamma}_1$ and $\widetilde{\gamma}_2$ have a *$\widetilde{\mathcal{F}}$-transverse intersection* at $\widetilde{\gamma}_1(t_1) = \widetilde{\gamma}_2(t_2)$, where $\widetilde{\gamma}_1 : J_1 \to \widetilde{\Sigma}$ and $\widetilde{\gamma}_2 : J_2 \to \widetilde{\Sigma}$ are lifts of γ_1 and γ_2 such that $\widetilde{\gamma}_1(t_1) = \widetilde{\gamma}_2(t_2)$. If $\gamma_1 = \gamma_2$ one speaks of a *$\mathcal{F}$-transverse self-intersection*. This means that if $\widetilde{\gamma}_1$ is a lift of γ_1, there exists $T \in \mathcal{G}$ such that $\widetilde{\gamma}_1$ and $T\widetilde{\gamma}_1$ have a $\widetilde{\mathcal{F}}$-transverse intersection at $\widetilde{\gamma}_1(t_1) = T\widetilde{\gamma}_1(t_2)$.

3.2 Recurrence, equivalence and accumulation

A transverse path (with respect to the foliation $\mathcal{F}$) $\gamma : \mathbb{R} \to \Sigma$ is *positively recurrent* if, for every $a < b$, there exist $c < d$, with $b < c$, such that $\gamma|_{[a,b]}$ and $\gamma|_{[c,d]}$ are equivalent. Similarly γ is *negatively recurrent* if, for every $a < b$, there exist $c < d$, with $d < a$, such that $\gamma|_{[a,b]}$ and $\gamma|_{[c,d]}$ are equivalent. Finally γ is *recurrent* if it is both positively and negatively recurrent.

Two transverse paths $\gamma_1 : \mathbb{R} \to \Sigma$ and $\gamma_2 : \mathbb{R} \to \Sigma$ are *equivalent at $+\infty$* if there exists a_1 and a_2 in $\mathbb{R}$ such that $\gamma_1|_{[a_1,+\infty)}$ and $\gamma_2|_{[a_2,+\infty)}$ are equivalent. Similarly γ_1 and γ_2 are *equivalent at $-\infty$* if there exists b_1 and b_2 in $\mathbb{R}$ such that $\gamma_1|_{(-\infty,b_1]}$ and $\gamma_2|_{(-\infty,b_2]}$ are equivalent.

A transverse path $\gamma_1 : \mathbb{R} \to \Sigma$ *accumulates positively* on the transverse path $\gamma_2 : \mathbb{R} \to \Sigma$ if there exist real numbers a_1 and $a_2 < b_2$ such that $\gamma_1|_{[a_1,+\infty)}$ and $\gamma_2|_{[a_2,b_2)}$ are equivalent. Similarly, γ_1 *accumulates negatively* on γ_2 if there exist real numbers b_1 and $a_2 < b_2$ such that $\gamma_1|_{(-\infty,b_1]}$ and $\gamma_2|_{(a_2,b_2]}$ are equivalent. Finally γ_1 *accumulates* on γ_2 if it accumulates positively or negatively on γ_2.

[1] In the whole text, "transverse" will mean "positively transverse".

3.3 Strips

We fix $T \in \mathcal{G} \setminus \{0\}$ and consider

- the annulus $\widehat{\Sigma} = \widetilde{\Sigma}/T$ (or, if $\widetilde{\Sigma}$ is not connected, the connected component of $\widetilde{\Sigma}/T$ that is an annulus);
- the covering projections $\pi : \widetilde{\Sigma} \to \widehat{\Sigma}$ and $\widehat{\pi} : \widehat{\Sigma} \to \Sigma$;
- the foliation $\widehat{\mathcal{F}}$ on $\widehat{\Sigma}$ induced by $\widetilde{\mathcal{F}}$.

Suppose that $\widehat{\Gamma}_*$ is a simple loop transverse to $\widehat{\mathcal{F}}$. Then, $\widehat{\Gamma}_*$ is essential and $\widetilde{\gamma}_* = \pi^{-1}(\widehat{\Gamma}_*)$ can be seen as an oriented line of $\widetilde{\Sigma}$, invariant by T and transverse to $\widetilde{\mathcal{F}}$. The set

$$\widehat{B} = \left\{ \widehat{z} \in \widehat{\Sigma} \,\middle|\, \widehat{\phi}_{\widehat{z}} \cap \widehat{\Gamma}_* \neq \emptyset \right\}$$

is an open annulus which is $\widehat{\mathcal{F}}$-saturated, meaning that it is a union of leaves. Similarly

$$\widetilde{B} = \pi^{-1}(\widehat{B}) = \left\{ \widetilde{z} \in \widetilde{\Sigma} \,\middle|\, \widetilde{\phi}_{\widetilde{z}} \cap \widetilde{\gamma}_* \neq \emptyset \right\}$$

is an $\widetilde{\mathcal{F}}$-saturated plane invariant by T. We will call such a set a *T-strip* (or a *strip* if we do not want to specify the covering transformation). In other words a T-strip is a $\widetilde{\mathcal{F}}$-saturated plane invariant by T defined as the union of the leaves that meet a transverse line invariant by T.

The frontier of $\widetilde{B}$, denoted $\partial\widetilde{B}$, is a union of leaves (possibly empty) and can be written $\partial\widetilde{B} = \partial\widetilde{B}^R \sqcup \partial\widetilde{B}^L$, where

$$\partial\widetilde{B}^R = \partial\widetilde{B} \cap R(\widetilde{\gamma}_*)\,, \qquad \partial\widetilde{B}^L = \partial\widetilde{B} \cap L(\widetilde{\gamma}_*).$$

Let us state some facts that can be proven easily (see [Lel23, Lemme 2.1.7]). Note first that:

- if there is a leaf $\widetilde{\phi} \subset \partial\widetilde{B}$ that is invariant by T, then the set $\partial\widetilde{B}^R$ or $\partial\widetilde{B}^L$ that contains $\widetilde{\phi}$ is reduced to this leaf;
- if $\widetilde{\gamma} : \mathbb{R} \to \widetilde{\Sigma}$ is transverse to $\widetilde{\mathcal{F}}$, then the set of real numbers t such that $\gamma(t) \in \widetilde{B}$ is an open interval (possibly empty).

Suppose now that $\widetilde{\gamma} : \mathbb{R} \to \widetilde{\Sigma}$ is transverse to $\widetilde{\mathcal{F}}$ and that

$$\left\{ t \in \mathbb{R} \,|\, \gamma(t) \in \widetilde{B} \right\} = (a, b),$$

where $-\infty \leq a < b \leq \infty$. Say that

- $\widetilde{\gamma}$ *draws* $\widetilde{B}$ if there exist $t < t'$ in (a, b) such that $\widetilde{\phi}_{\widetilde{\gamma}(t')} = T\widetilde{\phi}_{\widetilde{\gamma}(t))}$.

If, moreover, we suppose that $-\infty < a < b < +\infty$, say that:

- $\widetilde{\gamma}$ *crosses* $\widetilde{B}$ *from the right to the left* if $\widetilde{\gamma}(a) \in \partial\widetilde{B}^R$ and $\widetilde{\gamma}(b) \in \partial\widetilde{B}^L$;
- $\widetilde{\gamma}$ *crosses* $\widetilde{B}$ *from the left to the right* if $\widetilde{\gamma}(a) \in \partial\widetilde{B}^L$ and $\widetilde{\gamma}(b) \in \partial\widetilde{B}^R$;
- $\widetilde{\gamma}$ *visits* $\widetilde{B}$ *on the right* if $\widetilde{\gamma}(a) \in \partial\widetilde{B}^R$ and $\widetilde{\gamma}(b) \in \partial\widetilde{B}^R$;
- $\widetilde{\gamma}$ *visits* $\widetilde{B}$ *on the left* if $\widetilde{\gamma}(a) \in \partial\widetilde{B}^L$ and $\widetilde{\gamma}(b) \in \partial\widetilde{B}^L$.

We will say that $\widetilde{\gamma}$ *crosses* $\widetilde{B}$ if it crosses it from the right to the left or from the left to the right. Similarly, we will say that $\widetilde{\gamma}$ *visits* $\widetilde{B}$ if it visits it on the right or on the left. Note that $T(\widetilde{\gamma})$ satisfies the same properties as $\widetilde{\gamma}$. Note also that if $\widetilde{\gamma}$ visits $\widetilde{B}$ on the right, then $\partial\widetilde{B}^R$ is not reduced to a T-invariant leaf. Indeed, suppose that $\partial\widetilde{B}^R$ is reduced to a T-invariant leaf $\widetilde{\phi}$. With the notations above, we know that $\widetilde{\gamma}|_{(a,b)}$ is included in $L(\widetilde{\phi})$ because it is disjoint from $\widetilde{\phi}$ and $\gamma(a) \in \widetilde{\phi}$. Similarly, $\widetilde{\gamma}|_{(a,b)}$ is included in $R(\widetilde{\phi})$ because it is disjoint from $\widetilde{\phi}$ and $\gamma(b) \in \widetilde{\phi}$. We have found a contradiction. An analogous property holds if $\widetilde{\gamma}$ visits $\widetilde{B}$ on the left. Let us state a lemma whose proof is straightforward.

Lemma 3.1. *At least one of the following situations occurs (the two last assertions are not incompatible):*

- $\widetilde{\gamma}$ *crosses* $\widetilde{B}$*;*
- $\widetilde{\gamma}$ *visits* $\widetilde{B}$*;*

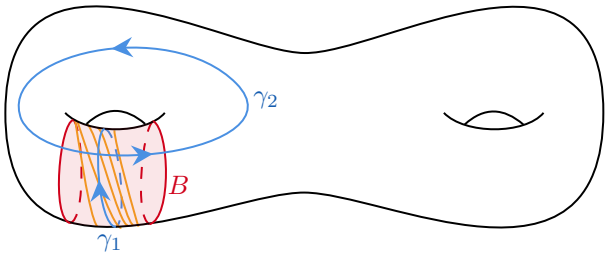


Figure 3.2: An example where Proposition 3.3 holds.

- *$\widetilde{\gamma}$ is equivalent to $\widetilde{\gamma}_*$ at $+\infty$ or at $-\infty$;*
- *$\widetilde{\gamma}$ accumulates on $\widetilde{\gamma}_*$ positively or negatively.*

Let us conclude this list of properties by the following ones (see [Lel23, Section 2.1.2.c]):

Proposition 3.2. *We have the following results:*

- *If $\widetilde{\gamma}$ visits and draws $\widetilde{B}$, then $\widetilde{\gamma}$ and $T(\widetilde{\gamma})$ have an $\widetilde{\mathcal{F}}$-transverse intersection and so $\gamma = \widetilde{\pi} \circ \widetilde{\gamma}$ has an $\mathcal{F}$-transverse self intersection.*
- *If $\widetilde{\gamma}_1$ crosses $\widetilde{B}$ from the right to the left, if $\widetilde{\gamma}_2$ crosses $\widetilde{B}$ from the left to the right and at least one of the paths $\widetilde{\gamma}_1$ or $\widetilde{\gamma}_2$ draws $\widetilde{B}$, then there exists $k \in \mathbb{Z}$ such that $\widetilde{\gamma}_1$ and $T^k(\widetilde{\gamma}_2)$ have an $\widetilde{\mathcal{F}}$-transverse intersection, and so $\gamma_1 = \widetilde{\pi} \circ \widetilde{\gamma}_1$ and $\gamma_2 = \widetilde{\pi} \circ \widetilde{\gamma}_2$ have an $\mathcal{F}$-transverse intersection.*

3.4 More about the accumulation property

In this final paragraph, we will suppose moreover than Σ is connected and that $\Sigma \neq \mathbb{R}^2/\mathbb{Z}^2$. The goal is to prove the following result that has its own interest and will be used in the sequel to prove Theorem 1.1. This statement is stronger than some results of [Lel23, Section 2.1.1]. It describes how a path γ_1 can accumulate on a path γ_2, the conclusion is that the situation depicted in Figure 3.2 is more or less the only one that can occur.

Proposition 3.3. *Suppose that $\gamma_1 : \mathbb{R} \to \Sigma$ is a positively recurrent transverse path that accumulates positively on a transverse path $\gamma_2 : \mathbb{R} \to \Sigma$. Then, there exists a transverse simple loop $\Gamma_* \subset \Sigma$ with the following properties.*

1. *The set B of leaves met by Γ_* is an open annulus of Σ.*
2. *The path γ_1 stays in B and is equivalent to the natural lift of Γ_*.*
3. *If $\widetilde{\gamma}_1$, $\widetilde{\gamma}_2$ are lifts of γ_1, γ_2 to the universal covering space $\widetilde{\Sigma}$ such that $\widetilde{\gamma}_1|_{[a_1,+\infty)}$ is equivalent to $\widetilde{\gamma}_2|_{[a_2,b_2)}$ and if $\widetilde{B}$ is the lift of B that contains $\widetilde{\gamma}_1$, then one of the inclusions $\widetilde{\phi}_{\widetilde{\gamma}_2(b_2)} \subset \partial\widetilde{B}^R$, $\widetilde{\phi}_{\widetilde{\gamma}_2(b_2)} \subset \partial\widetilde{B}^L$ holds. In the first case, we have $\widetilde{B} \subset L(\widetilde{\phi})$ for every $\widetilde{\phi} \subset \partial\widetilde{B}^R$ and in the second case, we have $\widetilde{B} \subset R(\widetilde{\phi})$ for every $\widetilde{\phi} \subset \partial\widetilde{B}^L$.*

Proof. Note first that if $\Gamma : \mathbb{T} \to \Sigma$ is a loop transverse to $\mathcal{F}$ and $\widetilde{\gamma} : \mathbb{R} \to \widetilde{\mathrm{dom}}(I)$ is a lift of Γ, there exists $T \in \mathcal{G}$ such that $\widetilde{\gamma}(t+1) = T(\widetilde{\gamma}(t))$ for every $t \in \mathbb{R}$. The foliation $\mathcal{F}$ being non singular, there is no transverse loop. We deduce that T is not the identity and that $\widetilde{\gamma}$ is simple. The periodicity condition tells us that $\widetilde{\gamma}$ is a line. So it defines naturally a T-strip, the union of leaves meeting $\widetilde{\gamma}$. We will say that T and $\widetilde{B}$ are associated to $\widetilde{\gamma}$.

Let us start with a lemma.

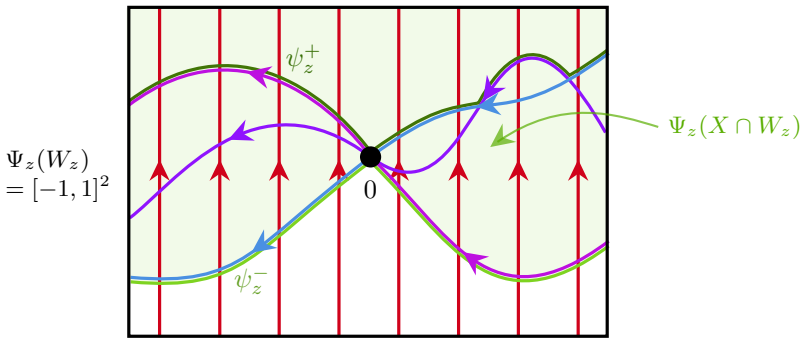


Figure 3.3: Local configuration of the path Γ and the foliation $\mathcal{F}$ (in red) around the point $0 = \Psi_z(z)$ in Sublemma 3.5.

Lemma 3.4. *Let $\Gamma : \mathbb{T} \to \Sigma$ be a transverse loop and $\widetilde{\gamma} : \mathbb{R} \to \widetilde{\mathrm{dom}}(I)$ a lift of Γ. Let $T \in \mathcal{G}$ be the deck transformation associated to $\widetilde{\gamma}$ and $\widetilde{B}$ the T-strip associated to $\widetilde{\gamma}$. Suppose that there exists a deck transformation $T' \in \mathcal{G}$ and $a \in \mathbb{R}$ such that $\widetilde{\gamma}|_{[a,a+1]}$ is equivalent to a subpath of $T'\widetilde{\gamma}$. Then $\widetilde{\gamma}|_{[a,a+1)} \cap T'\widetilde{\gamma} \neq \emptyset$. Moreover, if Γ is a simple loop, then $T' \in \langle T \rangle$.*

This lemma can be reduced easily to the following fact.

Sub-lemma 3.5. *Let $\Gamma : \mathbb{T} \to \Sigma$ be a loop transverse to $\mathcal{F}$. Then, there exists $z \in \Gamma$ and $z' \in \Gamma$ such that ϕ_z^+ does not meet Γ but at the end point and $\phi_{z'}^-$ does not meet Γ but at the end point.*

Proof of Lemma 3.4. By Sublemma 3.5, there exist z, z' in Γ (possibly equal) such that ϕ_z^+ and $\phi_{z'}^-$ do not meet Γ but at their end point. Denote $\widetilde{z} = \widetilde{\gamma}(b)$, $\widetilde{z}' = \widetilde{\gamma}(b')$ the respective lifts of z, z' that belong to $\widetilde{\gamma}|_{[a,a+1)}$. There is no loss of generality by supposing that $b \leq b'$. The union of $\widetilde{\phi}_{\widetilde{z}}^+$, $\widetilde{\phi}_{\widetilde{z}'}^-$ and $\widetilde{\gamma}([b,b'])$ defines a line of $\widetilde{\Sigma}$ that separates $\widetilde{\phi}_{\widetilde{z}}^- \setminus \{\widetilde{z}\}$ and $\widetilde{\phi}_{\widetilde{z}'}^+ \setminus \{\widetilde{z}'\}$. If $T'\widetilde{\gamma}$ does not meet $\widetilde{\gamma}([b,b'])$, then, by Sublemma 3.5, it does not meet this line. So (as by the hypothesis of the lemma it intersects both $\widetilde{\phi}_{\widetilde{z}}$ and $\widetilde{\phi}_{\widetilde{z}'}$) $T'\widetilde{\gamma}$ must intersect $\widetilde{\phi}_{\widetilde{z}}^- \setminus \{\widetilde{z}\}$ and $\widetilde{\phi}_{\widetilde{z}'}^+ \setminus \{\widetilde{z}'\}$, which contradicts the connectedness of $T'\widetilde{\gamma}$ and the fact that it is disjoint from the line. We have got a contradiction. We conclude that $T'\widetilde{\gamma} \cap \widetilde{\gamma}|_{[a,a+1)} \neq \emptyset$.

In the case where Γ is simple loop, we know that

$$T'\widetilde{\gamma} \cap \widetilde{\gamma} \neq \emptyset \iff T'\widetilde{\gamma} = \widetilde{\gamma} \iff T' \in \langle T \rangle. \qquad \square$$

Proof of Sublemma 3.5. Fix $z \in \Gamma$. The loop Γ being transverse to $\mathcal{F}$, there are finitely many parameters $t \in \mathbb{T}$ such that $z = \Gamma(t)$. Consequently, there exists a compact neighborhood W_z of z, a homeomorphism $\Phi_z : W_z \to [-1,1]^2$ and a finite set I_z such that:

- Φ_z sends z onto $(0,0)$;
- Φ_z sends $\mathcal{F}|_{W_z}$ onto the vertical foliation oriented upward;
- we have $\Phi_z(\Gamma \cap W_z) = \bigcup_{i \in I_z} \mathrm{gr}(\psi_{i,z})$, where $\psi_{i,z} : [-1,1] \to [-1,1]$ is a continuous function satisfying $\psi_{i,z}(0) = 0$.

Here the notation $\mathrm{gr}(\psi)$ denotes the graph of $\psi : [-1,1] \to [-1,1]$ oriented from the right to the left. See Figure 3.3 for an example of such a configuration.

Consider the two continuous functions

$$\psi_z^- = \min_{i \in I_z} \psi_{i,z}\,, \quad \psi_z^+ = \max_{i \in I_z} \psi_{i,z}$$

and define

$$\gamma_z^- = \Phi_z^{-1}(\mathrm{gr}(\psi_z^-))\,, \quad \gamma_z^+ = \Phi_z^{-1}(\mathrm{gr}(\psi_z^+)).$$

We will argue by contradiction by supposing that for any $z \in \Gamma$, the path ϕ_z^+ meets Γ in a point that is not the end point. In that case, for every $z \in \Gamma$, there exists a sub-path $\delta_z : [0,1] \to \Sigma$ of ϕ_z^+ such that

$$\delta_z(0) = z, \quad \delta_z(1) \in \Gamma, \quad \delta_z((0,1)) \cap \Gamma = \emptyset.$$

In particular we can define a first return map $\theta : \Gamma \to \Gamma$ by setting $\theta(z) = \delta_z(1)$. We will prove that $X = \bigcup_{z \in \Gamma} \delta_z([0,1])$ is a compact sub-surface (possibly with boundary).

Note that for every $z \in \Gamma$, the function θ induces a homeomorphism from a neighborhood of z in γ_z^+ to a neighborhood of $\theta(z)$ in $\gamma_{\theta(z)}^-$ and consequently that every point $\delta_z(t)$, $t \in (0,1)$, belongs to the interior of X. Moreover, reducing the W_z if necessary, one can suppose that θ is continuous on γ_z^+ and that $X_z = \bigcup_{z' \in \gamma_z^+} \delta_{z'}([0,1])$ is compact. The compact set

$$X'_z = \Phi_z^{-1}\Big(\big\{(x,y) \in [0,1]^2 \,\big|\, \psi_z^{-1}(x) \leq y \leq \psi_z^+(x)\big\}\Big)$$

satisfies the equality

$$X'_z \setminus \{z\} = \bigcup_{z' \in (W_z \cap \Gamma) \setminus \gamma_z^+} \delta_{z'}([0,1]),$$

which implies that

$$\bigcup_{z' \in W_z \cap \Gamma} \delta_{z'}([0,1]) = X_z \cup X'_z.$$

The loop Γ can be covered by finitely many W_z, $z \in \Gamma$, and so X is compact. By compactness of X, reducing the W_z if necessary, one can suppose that the following alternative holds:

1. either γ_z^- is contained in the image of θ and W_z is included in X,
2. or γ_z^- is disjoint from the image of θ and we have

$$X \cap W_z = \Phi_z^{-1}\Big(\big\{(x,y) \in [0,1]^2 \,\big|\, \psi_z^{-1}(x) \leq y \leq 1\big\}\Big).$$

The fact that every point $\delta_z(t)$, $z \in \Gamma$, $t \in (0,1)$, belongs to the interior of X implies that every point z of the frontier of X belongs to Γ. Moreover W_z satisfies the second property. We have proved that X is a compact sub-surface (possibly with boundary). Note that for every $z \in \partial X$ it holds that $\phi_z^+ \setminus \{z\} \subset \mathrm{int}(X)$ (in other terms the foliation is pointing inward on the boundary).

By hypothesis, Σ is connected and different from $\mathbb{R}^2/\mathbb{Z}^2$. So, it does not bear a non-singular foliation. We deduce that X is a surface with boundary. More precisely it is homeomorphic to the closed annulus because it bears a non singular foliation. Let $\Psi : X \to \mathbb{S}^2$ be a topological embedding compatible with the usual orientations. The loop $\Psi(\Gamma)$ is homologous to 0 in $\mathbb{S}^2$ and one can define a dual function $\eta : \mathbb{S}^2 \setminus \Psi(\Gamma) \to \mathbb{Z}$. Such a function is defined by the following property: for every z, z' in $\mathbb{S}^2 \setminus \Psi(\Gamma)$ and every path β joining z to z', the algebraic intersection number $\Psi(\Gamma) \wedge \beta$ is equal to $\eta(z') - \eta(z)$. Let U be a connected component of $\mathbb{S}^2 \setminus \Psi(\Gamma)$ where η reaches its maximum. The set $\Psi(\Gamma)$ being connected, the closure of U is a topological disk. Moreover the fact that η reaches its maximum in U implies that for every $z \in \partial U$ it holds that $\phi_z^+ \setminus \{z\} \subset U$. So U is not a connected component of $\mathbb{S}^2 \setminus \Psi(X)$ and it holds that $\overline{U} \subset \psi(X)$. Summarizing, we have found a closed topological disk bearing a non-singular foliation pointing inward on the boundary. We have got a contradiction. □

Let us explain how to construct the simple loop Γ_*, associated to γ_1, that appears in Proposition 3.3. As γ_1 is positively recurrent, there exist two numbers $c_1 < c'_1$, with $c_1 > a_1$, such that $\phi_{\gamma_1(c_1)} = \phi_{\gamma_1(c'_1)}$ (see Figure 3.4 for these different points). It implies that $\gamma_1|_{[c_1,c'_1]}$ is equivalent to a transverse path $\gamma_* : [c_1, c'_1] \to \Sigma$ such that $\gamma_*(c_1) = \gamma_*(c'_1)$. To get γ_*, one can modify γ_1 on a left neighborhood of c'_1, the modification taking place in a flow box containing the segment of leaf joining $\gamma_1(c_1)$ to $\gamma_1(c'_1)$. The set

$$X = \big\{(t,t') \in [c_1,c'_1]^2 \,\big|\; t < t' \text{ and } \gamma_*(t) = \gamma_*(t')\big\}$$

is non empty (because it contains (c_1, c'_1)) and compact. Indeed, it is closed in $\{(t,t') \in [c_1,c'_1]^2 \mid t < t'\}$, an its closure in the compact set $\{(t,t') \in [c_1,c'_1]^2 \mid t \leq t'\}$ does not contain any couple (t,t). The function $(t,t') \mapsto t' - t$ being continuous and positive on X, reaches its minimum at a couple (c''_1, c'''_1). So, replacing (c_1, c'_1) with (c''_1, c'''_1) if necessary, one can always suppose that $\gamma_*(t) \neq \gamma_*(t')$ if $t < t'$ and $(t,t') \neq (c_1, c'_1)$. One defines a continuous map $\Gamma_* : \mathbb{T} \to \mathbb{S}$ by setting $\Gamma_*(t + \mathbb{Z}) = \gamma_*((1-t)c_1 + tc'_1)$ if $t \in [0,1]$. This map being injective, it is a simple loop transverse to $\mathcal{F}$. We denote B the union of leaves met by Γ_*.

By hypothesis there exist two lifts $\widetilde{\gamma}_1$ and $\widetilde{\gamma}_2$ of respectively γ_1 and γ_2 to $\widetilde{\Sigma}$ such that $\widetilde{\gamma}_1|_{[a_1,+\infty)}$ and $\widetilde{\gamma}_2|_{[a_2,b_2)}$ are equivalent. We denote $\widetilde{\gamma}_*$ the lift of Γ_* that contains $\widetilde{\gamma}_1(c_1)$. Then we consider the deck transformation T and the T-strip $\widetilde{B}$ associated to $\widetilde{\gamma}_*$. Note that T is primitive because Γ_* is simple.

Lemma 3.6. *The path $\widetilde{\gamma}_1|_{[c_1,+\infty)}$ is included in $\widetilde{B}$.*

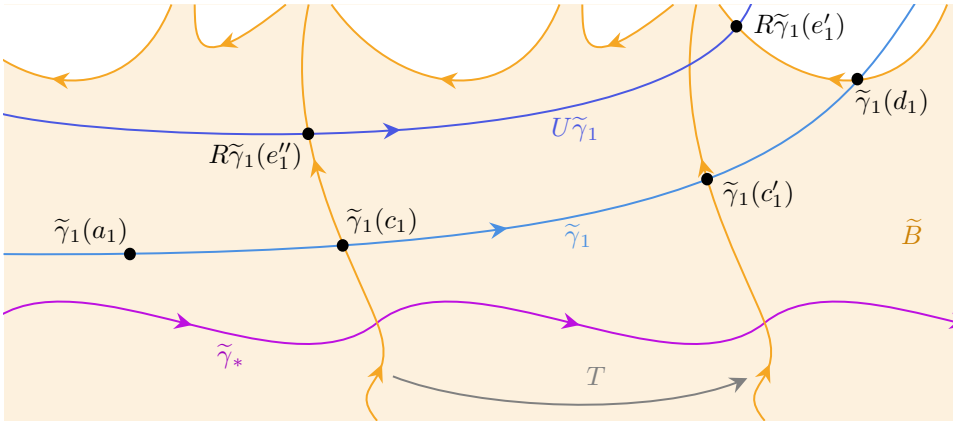


Figure 3.4: The different objects appearing in the proof of Proposition 3.3, Lemma 3.6 and Claim 3.7. The leaves are in orange.

Proof. We will argue by contradiction and suppose it is not. Then there exists $d_1 > c_1'$, uniquely defined, such that $\widetilde{\gamma}_1(d_1) \notin \widetilde{B}$ and $\widetilde{\gamma}_1|_{[c_1,d_1)} \subset \widetilde{B}$ (see Figure 3.4).

Claim 3.7. *There exists a deck transformation $R \in \mathcal{G}$ and real numbers $e_1 < e_1'$, with $e_1 \geq a_1$, such that either $R\widetilde{\gamma}_1|_{[e_1,e_1']}$ draws and crosses $\widetilde{B}$, or it draws and visits $\widetilde{B}$.*

Proof. Note that to prove this claim one has to find $R \in \mathcal{G}$ and $e_1 < e_1'$ such that $R\widetilde{\gamma}_1|_{[e_1,e_1']}$ draws $\widetilde{B}$ and both $R\widetilde{\gamma}_1(e_1)$ and $R\widetilde{\gamma}_1(e_1')$ do not belong to $\widetilde{B}$.

As γ_1 is positively recurrent, there exist real numbers $e_1'' < e_1'$, with $e_1'' > d_1$, and a deck transformation $R \in \mathcal{G}$ such that $R\widetilde{\gamma}_1|_{[e_1'',e_1']}$ is equivalent to $\widetilde{\gamma}_1|_{[c_1,d_1]}$; in particular:

- $\widetilde{\gamma}_1|_{[c_1,c_1']}$ is equivalent to a subpath of $R\widetilde{\gamma}_1|_{[e_1'',e_1']}$;
- $R\widetilde{\gamma}_1([e_1'',e_1')) \subset \widetilde{B}$ and $R\widetilde{\gamma}_1(e_1') \notin \widetilde{B}$.

To prove the claim, it is sufficient to show that $R\widetilde{\gamma}_1([a_1,e_1')) \not\subset \widetilde{B}$, because in that case there exists $e_1 \in [a_1,e_1'']$ such that $R\widetilde{\gamma}_1((e_1,e_1')) \subset \widetilde{B}$ and $R\widetilde{\gamma}_1(e_1) \notin \widetilde{B}$.

We argue by contradiction. Suppose that $R\widetilde{\gamma}_1([a_1,e_1'))$ is contained in $\widetilde{B}$. Then $\widetilde{\gamma}_1([a_1,e_1'))$ is contained in $R^{-1}(\widetilde{B})$. Recall that there exists t such that $\widetilde{\gamma}_*|_{[t,t+1]}$ is equivalent to $\widetilde{\gamma}_1|_{[c_1,c_1']}$ which is a subpath of $\widetilde{\gamma}_1|_{[a_1,e_1')}$. It implies that $\widetilde{\gamma}_*|_{[t,t+1]}$ is equivalent to a subpath of $R^{-1}\widetilde{\gamma}_*$ because $\widetilde{\gamma}_1([a_1,e_1'))$ is contained in $R^{-1}(\widetilde{B})$. Lemma 3.4 applies and ensures that $R^{-1} \in \langle T \rangle$. As $\widetilde{B}$ is invariant by T, the condition $R\widetilde{\gamma}_1([a_1,e_1')) \subset \widetilde{B}$ gives $\widetilde{\gamma}_1([a_1,e_1')) \subset \widetilde{B}$. This contradicts the condition $\widetilde{\gamma}_1(d_1) \notin \widetilde{B}$, because $a_1 < d_1 < e_1'$. □

As γ_1 is positively recurrent, there exist sequences $(e_{1,n})_{n\geq 0}$ and $(e_{1,n}')_{n\geq 0}$ with $a_1 < e_{1,n} < e_{1,n}' < e_{1,n+1}$, and a sequence $(R_n)_{n\geq 0}$ of deck transformations, such that $R_n\widetilde{\gamma}_1|_{[e_{1,n},e_{1,n}']}$ is equivalent to $R\widetilde{\gamma}_1|_{[e_1,e_1']}$. As $\widetilde{\gamma}_1$ accumulates on $\widetilde{\gamma}_2$, a similar statement holds for $\widetilde{\gamma}_2$: there exist sequences $(e_{2,n})_{n\geq 0}$ and $(e_{2,n}')_{n\geq 0}$ with $a_2 < e_{2,n} < e_{2,n}' < e_{2,n+1} < b_2$ such that $R_n\widetilde{\gamma}_2|_{[e_{2,n},e_{2,n}']}$, is equivalent to $R\widetilde{\gamma}_1|_{[e_1,e_1']}$. Note that the R_n are all different because every leaf of $\widetilde{\mathcal{F}}$ intersects $\widetilde{\gamma}_2([a_2,b_2])$ at most once.

We have two possibilities given by Claim 3.7: either $R\widetilde{\gamma}_1|_{[e_1,e_1']}$ draws and crosses $\widetilde{B}$, or it draws and visits $\widetilde{B}$.

Suppose that $R\widetilde{\gamma}_1|_{[e_1,e_1']}$ draws and crosses $\widetilde{B}$. In this case, for any $n \in \mathbb{N}$, the path $R_n\widetilde{\gamma}_2|_{[e_{2,n},e_{2,n}']}$ intersects $\widetilde{\gamma}_*$. Replacing R_n with $T^{k_n} \circ R_n$ for a certain $k_n \in \mathbb{Z}$ if necessary, one can suppose that $R_n\widetilde{\gamma}_2|_{[e_{2,n},e_{2,n}']}$ intersects $\widetilde{\gamma}_*|_{[t,t+1]}$ and so $R_n^{-1}(\widetilde{\gamma}_*|_{[t,t+1]})$ intersects $\widetilde{\gamma}_2([a_2,b_2])$. It contradicts the fact that the action of $\mathcal{G}$ on compact subsets is proper.

Suppose now that $R\widetilde{\gamma}_1|_{[e_1,e_1']}$ draws and visits $\widetilde{B}$. Then $R\widetilde{\gamma}_1|_{[e_1,e_1']}$ and $TR\widetilde{\gamma}_1|_{[e_1,e_1']}$ have an $\widetilde{\mathcal{F}}$-transverse intersection. One deduces that for any $n \in \mathbb{N}$, one has $R_n\widetilde{\gamma}_2|_{[e_{2,n},e_{2,n}']}$ and $TR\widetilde{\gamma}_1|_{[e_1,e_1']}$ have an $\widetilde{\mathcal{F}}$-transverse intersection because $R_n\widetilde{\gamma}_2|_{[e_{2,n},e_{2,n}']}$ and $R\widetilde{\gamma}_1|_{[e_1,e_1']}$ are equivalent. Consequently, it holds that $R_n\widetilde{\gamma}_2|_{[e_{2,n},e_{2,n}']} \cap TR\widetilde{\gamma}_1|_{[e_1,e_1']}$

$\neq \emptyset$ and so that $R_n\widetilde{\gamma}_2|_{[a_2,b_2]} \cap TR\widetilde{\gamma}_1|_{[e_1,e_1']} \neq \emptyset$. It contradicts once again the fact that the action of $\mathcal{G}$ on compact subsets is proper. This finishes the proof of Lemma 3.6. □

By Lemma 3.6, we know that $\widetilde{\gamma}_1|_{[c_1,+\infty)}$ stays in $\widetilde{B}$. We first prove that $\widetilde{\gamma}_1$ cannot accumulate in $\widetilde{\gamma}_*$.

Indeed, otherwise, as γ_1 is positively recurrent, there exist deck transformations $(R_n)_{n\geq 0} \in \mathcal{G}$ and parameters $d_n < d_n'$ both going to $+\infty$ such that $\widetilde{\gamma}_1|_{[d_n,d_n']}$ is equivalent to $R_n\widetilde{\gamma}_1|_{[c_1,c_1']}$, which is itself equivalent to $R_n\widetilde{\gamma}_*|_{[t,t+1]}$. The fact that $\widetilde{\gamma}_1$ accumulates in $\widetilde{\gamma}_*$ implies that $R_n \notin \langle T\rangle$ eventually. Recall that for any n, the path $\widetilde{\gamma}_1|_{[d_n,d_n']}$ is equivalent to a subpath of $\widetilde{\gamma}_*$; this allows to apply Lemma 3.4 to the simple path Γ_*, which implies that $R_n \in \langle T\rangle$, a contradiction.

Hence, there exists $t_1 \in \mathbb{R}$ such that $\widetilde{\gamma}_1|_{[c_1,+\infty)}$ is equivalent to $\widetilde{\gamma}_*|_{[t_1,+\infty)}$. Moreover it is equivalent to $\widetilde{\gamma}_2|_{[c_2,b_2)}$, where $c_2 \in [a_2,b_2]$. It implies that $\widetilde{\phi}_{\widetilde{\gamma}_2(b_2)} \subset \partial\widetilde{B}$. We do not lose generality by supposing that $\widetilde{\phi}_{\widetilde{\gamma}_2(b_2)} \subset \partial\widetilde{B}^L$. We choose $a_2' \in [c_2,b_2)$ such that $\widetilde{\gamma}_2([a_2',b_2]) \in L(\widetilde{\gamma}_*)$.

Lemma 3.8. *For every leaf $\widetilde{\phi} \subset \partial\widetilde{B}^L$ it holds that $\widetilde{B} \subset R(\widetilde{\phi})$.*

Proof. See Figure 3.5 for an example of configuration of the proof. Suppose that there exists a leaf $\widetilde{\phi}_0 \subset \partial\widetilde{B}^L$ such that $\widetilde{B} \subset L(\widetilde{\phi})$. One can find a transverse path $\widetilde{\gamma}_3 : [a_3,b_3] \to \widetilde{\Sigma}$ such that $\widetilde{\gamma}_3(a_3) \in \widetilde{\phi}_0$ and $\widetilde{\gamma}_3((a_3,b_3]) \subset \widetilde{B}$. Such a path enters in $\widetilde{B}$ by the left. By taking a smaller b_3 if necessary, we can suppose moreover that $\widetilde{\gamma}_3([a_3,b_3]) \subset L(\widetilde{\gamma}_*)$. We will prove that it prevents $\widetilde{\gamma}_1$ accumulating positively in $\widetilde{\gamma}_2$. If $\widetilde{\lambda}$ is an oriented line of $\widetilde{B}$, denote $R_{\widetilde{B}}(\widetilde{\lambda})$ the connected component of $\widetilde{B} \setminus \widetilde{\lambda}$ located on the right of $\widetilde{\lambda}$ and $L_{\widetilde{B}}(\widetilde{\lambda})$ the connected component of $\widetilde{B} \setminus \widetilde{\lambda}$ located on the left of $\widetilde{\lambda}$. One defines two oriented lines $\widetilde{\lambda}_2$, $\widetilde{\lambda}_3$ of $\widetilde{B}$ by setting

$$\widetilde{\lambda}_2 = \big(\widetilde{\gamma}_2|_{[a_2',b_2)}\big)^{-1}\widetilde{\phi}^+_{\widetilde{\gamma}_2(a_2')}, \qquad \widetilde{\lambda}_3 = \widetilde{\gamma}_3|_{(a_3,b_3]}\,\widetilde{\phi}^+_{\widetilde{\gamma}_3(b_3)}. \tag{3.1}$$

The line $\widetilde{\gamma}_*$ intersects $\widetilde{\phi}_{\widetilde{\gamma}_2(a_2')}$ in a unique point $\widetilde{z}_2$ and we have $\widetilde{z}_2 \in \widetilde{\phi}^+_{\widetilde{\gamma}_2(a_2')}$. Similarly, $\widetilde{\gamma}_*$ intersects $\widetilde{\phi}_{\widetilde{\gamma}_3(b_3)}$ in a unique point $\widetilde{z}_3$ and we have $\widetilde{z}_3 \in \widetilde{\phi}^+_{\widetilde{\gamma}_3(b_3)}$. Denote $\widetilde{\sigma}_2 \subset \widetilde{\phi}_{\widetilde{\gamma}_2(a_2)}$ the segment that joins $\widetilde{\gamma}_2(a_2')$ to $\widetilde{z}_2$ and $\widetilde{\sigma}_3 \subset \widetilde{\phi}_{\widetilde{\gamma}(b_3)}$ the segment that joins $\widetilde{\gamma}_3(b_3)$ to $\widetilde{z}_3$. By compactness of all segments, if n is large enough, then we have

$$T^n\big(\widetilde{\gamma}_3([a_3,b_3]) \cup \widetilde{\sigma}_3\big) \cap \big(\widetilde{\gamma}_2([a_2',b_2]) \cup \widetilde{\sigma}_2\big) = \emptyset. \tag{3.2}$$

Moreover, one can suppose that

$$T^n\widetilde{\phi}_{\widetilde{\gamma}_3(b_3)} \subset L\big(\widetilde{\phi}_{\widetilde{\gamma}_2(a_2')}\big).$$

The fact that $\widetilde{\gamma}_2([a_2',b_2])$, $\widetilde{\sigma}_2$, $\widetilde{\gamma}_3([a_3,b_3])$ and $\widetilde{\sigma}_3$ are included in $L(\widetilde{\gamma}_*)$ while $\widetilde{\phi}^+_{\widetilde{z}_2}$ and $\widetilde{\phi}^+_{\widetilde{z}_3}$ are included in $\overline{R(\widetilde{\gamma}_*)}$ tells us that

$$T^n\widetilde{\phi}^+_{\widetilde{z}_3} \cap (\widetilde{\gamma}_2([a_2',b_2]) \cup \widetilde{\sigma}_2) = T^n\left(\widetilde{\gamma}_3([a_3,b_3]) \cup \widetilde{\sigma}_3\right) \cap \widetilde{\phi}^+_{\widetilde{z}_2} = \emptyset. \tag{3.3}$$

By combining (3.1), (3.2) and (3.3), we deduce that the lines $\widetilde{\lambda}_2$ and $T^n\widetilde{\lambda}_3$ are disjoint.

The sub-path of $\widetilde{\gamma}_*$ that joins $\widetilde{z}_2$ to $T^n\widetilde{z}_3$ is disjoint from $\widetilde{\lambda}_2$ and $T^n\widetilde{\lambda}_3$ but at its endpoints, entering in $L_{\widetilde{B}}(\widetilde{\lambda}_2)$ at $\widetilde{z}_2$ and leaving $R_{\widetilde{B}}(T^n\widetilde{\lambda}_3)$ at $T^n\widetilde{z}_3$. Consequently the inclusion $\overline{L_{\widetilde{B}}(T^n\widetilde{\lambda}_3)} \subset L_{\widetilde{B}}(\widetilde{\lambda}_2)$ holds. Every leaf $\widetilde{\phi} \subset L_{\widetilde{B}}(\widetilde{\phi}_{T^n\widetilde{z}_3})$ is disjoint from $T^n\widetilde{\lambda}_3$. It is contained in $L(T^n\widetilde{\lambda}_3)$ because the sub-path of $\widetilde{\gamma}_*$ that joins $\widetilde{\phi}_{T^n(\widetilde{z}_3)}$ to $\widetilde{\phi}$ is disjoint from $T^n\widetilde{\lambda}_3$ but at $T^n\widetilde{z}_3$ and enters in $L_{\widetilde{B}}(T^n\widetilde{\lambda}_3)$ at $T^n\widetilde{z}_3$. The contradiction comes from the fact that $\widetilde{\phi}$ must intersect $\widetilde{\gamma}_2|_{[a_2',b_2)}$ because $\widetilde{\phi} \subset L_{\widetilde{B}}(\widetilde{\phi}_{\widetilde{z}_2})$. □

Lemma 3.9. *The set B is an open annulus of Σ.*

Proof. Suppose it is not. Then there exists a deck transformation $T' \notin \langle T\rangle$ of $\widetilde{\Sigma}$ such that $T'\widetilde{B} \cap \widetilde{B} \neq \emptyset$. As $\widetilde{B}$ is the set of leaves met by $\widetilde{\gamma}_*$, it implies the existence of $t \in \mathbb{R}$ such that $T'\widetilde{\gamma}_*(t) \in \widetilde{B}$. The line $\widetilde{\gamma}_*$ lifts the simple loop Γ_* and so we have $T'\widetilde{\gamma}_* \cap \widetilde{\gamma}_* = \emptyset$. Moreover, there is at least one leaf of $\widetilde{\mathcal{F}}$ that is met both by $\widetilde{\gamma}_*$ and $T'\widetilde{\gamma}_*$. Consequently, one of the following inclusions $\overline{L(T'\widetilde{\gamma}_*)} \subset L(\widetilde{\gamma}_*)$, $\overline{L(\widetilde{\gamma}_*)} \subset L(T'\widetilde{\gamma}_*)$ holds. Replacing T' by T'^{-1} if necessary, one can suppose that the first inclusion holds, which implies that $T'\widetilde{\gamma}_* \subset L(\widetilde{\gamma}_*)$.

Note that $T'\widetilde{\gamma}_*$ cannot accumulate on $\widetilde{\gamma}_*$ (neither positively nor negatively) because the natural lift γ_* of Γ_* is recurrent and so, by Lemma 3.4, cannot accumulate on itself. Moreover it cannot be equivalent to $\widetilde{\gamma}_*$ neither at $+\infty$ nor at $-\infty$ (by using Lemma 3.4). It cannot cross $\widetilde{B}$ because $T'\widetilde{\gamma}_* \cap \widetilde{\gamma}_* = \emptyset$. It remains to prove that it cannot visit $\widetilde{B}$.

Using the fact that $T'\widetilde{\gamma}_* \subset L(\widetilde{\gamma}_*)$, the line $T'\widetilde{\gamma}_*$ must visit $\widetilde{B}$ by the left if it visits $\widetilde{B}$. This contradicts Lemma 3.8: no transverse trajectory enters in $\widetilde{B}$ by the left side. □

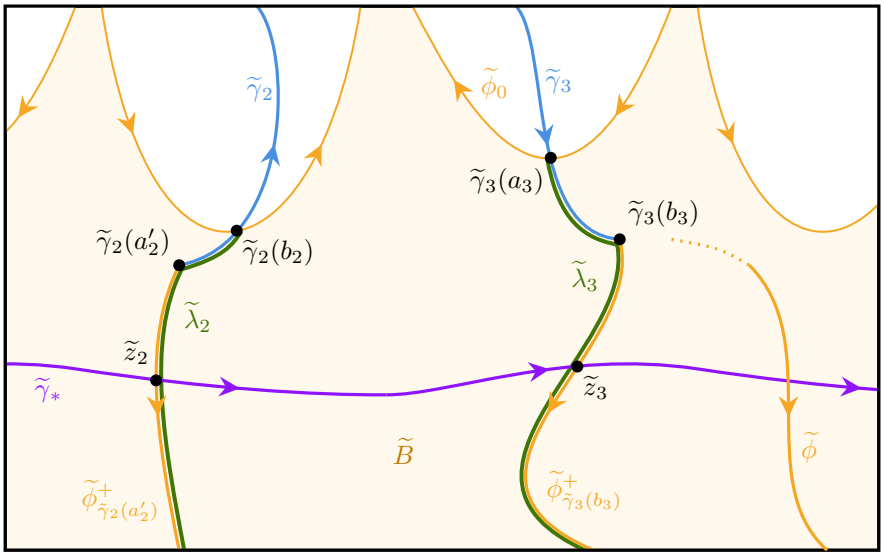


Figure 3.5: The configuration of the proof of Lemma 3.8.

To prove Proposition 3.3, it remains to prove that γ_1 is entirely contained in B (which will imply that $\widetilde{\gamma}_1$ is entirely contained in $\widetilde{B}$). But this is implied by the facts that $\gamma_1|_{[a_1,+\infty)}$ is contained in B and that γ_1 is recurrent. This finishes the proof of Proposition 3.3. □

References

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